BME 4232

AI-Driven Glucose Monitoring System for Telemedicine Application



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Abstract

This project is basically focusing on developing a low cost, AI driven glucose concentration estimation system that can be an alternative option to conventional costly spectrophotometric procedure in resource constraining areas. This system utilizes the phototransistors that are being incorporated in the sensing mechanism to measure the glucose level of the solutions and ensure real time data acquisition from the user. For data acquisition and visualization, python has been used due to which we will be able to gain the absorption spectra from which we can interpret the glucose concentration level. The collected data is organized into a structured dataset. Random Forest ML algorithm has been employed to analyze the dataset and enable an accurate prediction of glucose levels afterwards. This trained machine learning model has been integrated in the system so that it can facilitate quick estimation of glucose concentration of a patient. In addition, a userfriendly application has also been developed for tracking and visualizing the data so that the user and healthcare professionals can monitor and analyze the data. The proposed spectrophotometer can be considered as a basic prototype of device that can be handy in educational purposes, preliminary research and also in resource limited environments. As it is being considered as a prototype a huge amount of work can be done over this for improving its performance.

Chapter 1

Introduction

1.1 Overview

Diabetes is one of the most common lifelong chronic diseases in human beings. It is mainly caused by genetic factors, immune disorders and other factors on the human body, leading to islet function decline and insulin resistance, resulting in imbalance of glucose level in the body, which is manifested as failure of glucose metabolism and hyperglycemia [1]. A plethora of biosensors have been developed to provide diagnostic information regarding a patient's health status. Many different types of sensors have been investigated, and a 2010 review by Toghill and Compton provides a great insight into enzymatic and non-enzymatic electrochemical glucose sensing approaches studied over the past decade [2]. Spectroscopic methods for non-invasive glucose detection have also been growing in popularity, with Raman and infrared spectroscopy being the focus of particular attention. However, the real challenge that remains is the creation of biosensors for daily use by patients in personalised monitoring.

Conventional glucose monitoring systems and laboratory based techniques are often costly, require specialized equipment and may not be easily accessible in resource constrained or remote areas.

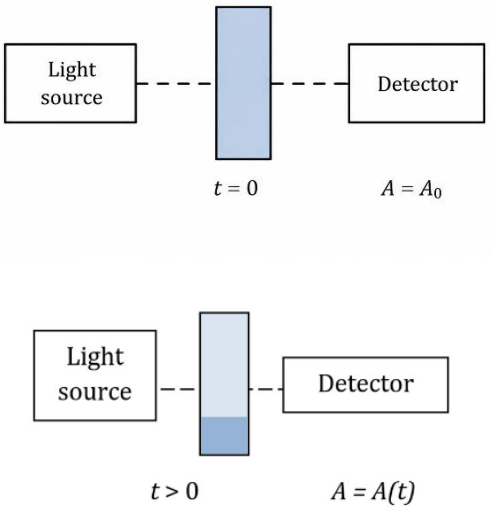


Figure 1.1. The principle of the spectrophotometric measurement [3].

1.2 Minimally invasive Glucose Monitoring

Minimally invasive blood glucose concentration detection, as its name implies, refers to the detection of human blood glucose with minimal penetration of the skin, causing very little damage to human tissue. There are lots of methods for partially-invasive blood glucose detection, which can be generally divided into micro needle based methods, minimally invasive electrochemical sensor and implantable sensor methods. Among various methods, spectrometry based method can also be integrated in a partially invasive method.

A spectrophotometer is a powerful analytical instrument widely used in science and industry to measure the intensity of light absorbed or transmitted by a sample at specific wavelengths. It operates by passing a beam of light through a sample and detecting the light that emerges on the other side[4]. By analyzing the interaction between light and the molecules in a sample, the spectrophotometer provides valuable data about the sample's composition and concentration. This data is typically represented as a spectrophotogram, a graph showing absorbance as a function of wavelength [5]. The working principle of a spectrophotometer is rooted in Beer-Lambert's law, which states that the absorbance of light is directly proportional to the concentration of the absorbing species and the path length of the light through the sample. By comparing the transmitted light intensity (I) with the initial light intensity (I0), the absorbance is calculated using [5].

$$A = -log\left(\frac{I}{I_0}\right) \quad (1)$$

This process allows for precise quantification of substances in a solution.

For this work, an array of LEDs are used which generates light at different wavelengths. The measured intensity differences caused by the glucose concentration. Its major advantages are it is partially invasive, easy to carry and helpful for rural areas. Furthermore, it provides with prediction of glucose level accurately.

1.3 AI driven Approach

Machine learning techniques are widely used for predicting the performance of a device or a system. By learning patterns, ML models can accurately predict the outcomes. In engineering and sensing applications, ML has been employed to predict the performance of different sensing devices. For this work, ML is used for predicting the glucose concentration in real time.

Artificial intelligence (AI)–driven glucose concentration predicting systems are emerging telemedicine tools, which allows accurate measurement of blood glucose data for better diabetes management and remote clinical decision support. These predictions rely on supervised learning models such as support vector machines, particularly random forests, which are well-suited for tabular data [6]. By incorporating contextual awareness, AI models offer not just numerical predictions but also risk scores, helping patients understand the likelihood and severity of impending glucose excursions[7].

In this work, at first a dataset will be developed from the glucose related measurements that mainly is the concentration of the glucose. Then ML will use this dataset to understand the pattern between the data and improve the accuracy of the prediction overtime. Later on, application will be developed. This developed application allows the monitoring of the glucose data, health status and also allows healthcare professionals to monitor these data from afar and give directions accordingly.

Chapter 2

Literature Review

In recent years, significant efforts have been made to develop low-cost and portable spectrophotometers that provide reliable performance for various applications, including glucose monitoring. Many of these works leverage microcontrollers, such as Arduino and affordable optical components to create accessible and easy-to-use systems. Some have added AI applications also. Table 2.1 summarizes the existing literature related to Arduino-based spectrophotometric systems, AI-driven glucose monitoring technologies and telemedicine-integrated healthcare applications.

Table 2.1: Literature review table

Ref.	Study	Methodology	Key Features	Limitations
[8]	Arduino-based Spectrophotometer	Developed a low-cost spectrophotometer for concentration measurement of different solutions	Cost effective, easy fabrication and suitable for small scale applications	No AI integration
[9]	Portable Photometer and Fluorimeter	Developed a portable Arduino based photometer and fluorimeter system	Portable, effective protein and phosphate ion measurement	Lesser healthcare applications
[10]	Mobile Application-Based Glucose Monitoring System	Developed a glucose monitoring system integrated with a mobile application and AI techniques	Real time monitoring, mobile accessibility and smart healthcare support	Limited discussion on wavelength-based sensing and explainable AI analysis
[11]	AI-Driven Glucose Monitoring and Telemedicine System	Developed a sensor-based glucose monitoring system using AI prediction models and telemedicine integration	Automated glucose prediction, remote monitoring and telemedicine support	Limited explainability and feature-level interpretation of prediction results

Chapter 3

Objectives

This work seeks to develop an efficient, cost effective and intelligent glucose concentration estimation system. The system can provide accurate measurement of glucose data. The main objectives are as follows:

- To design and develop a minimally invasive glucose estimation system using spectrophotometric techniques.
- To collect and pre-process the data from the setup and train a machine learning model for trustworthy prediction of glucose levels.
- To develop a telemedicine application for estimating glucose levels and make it accessible for healthcare professionals and other caregivers.

Chapter 4

Methodology

4.1 Theoretical Framework

Theoretical framework for this project consists of the fundamental concept of Beer-Lambert law. Mainly in theoretical approach the genuine spectrophotometer works on the fundamental concept of Beer-Lambert Law indicating the fact that the absorbance of the light is directly proportional to the concentration of the species and the path length through the solution.

$$\text{Absorbance, } A = \epsilon l C \tag{2}$$

Here, ε= Molar absorptivity of the solution, l =length, C = Concentration

But in this project the logarithmic format of Beer-Lambert law has been incorporated for just detecting whether a specific component is being present or not within the solution.

$$A = -log\left(\frac{I}{I_0}\right) \tag{3}$$

According to the intensity level the used phototransistors will generate a specified amount of current. That analog signal is being fed to the Arduino and hence afterwards processing that data we will be able to get an ultimate result based on the absorbance and wavelength level.

4.2 Working procedure:

The working procedure of the project is divided in different phases where each phases contributes in a different way. The stages include spectrophotometric sensing, dataset creation and intelligent analysis. The overall working procedure is visualized in the following flowchart in figure 4.1.

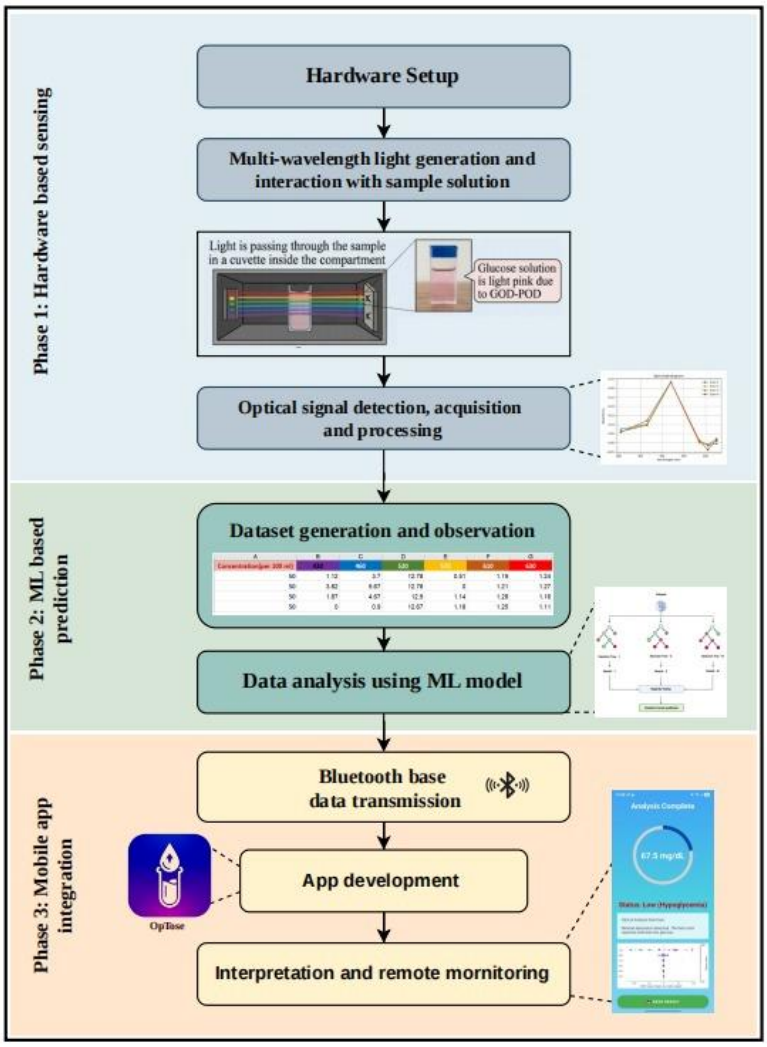


Figure 4.1. Representation of overall work flowchart

4.3 Hardware Setup:

- **LED and Phototransistor holder**

For gaining an array of LEDs, we designed a holder that will consist of an array of LEDs within it. Similarly, for ensuring proper alignment the array of sensor will also be incorporated within a holder. In the case of LED holder, the height of it will be like 54.5mm and width will be 26mm as demonstrated in figure 4.2. The thickness of it will be 8.2 mm. 6 holes will be there

maintaining gap of 8.6mm. The first hole from the ground will maintain a distance of 6.5mm. The diameter of the hole is equivalent to 5.2mm. Some perpendicular cuts are also being present for shouldering purpose maintain a depth of 2mm and a width of 1.5mm. The LED holder will contain the LEDs in a specific order of wavelengths. Such as violet will be in the lowest cut position. Then blue, green, yellow, orange and red.

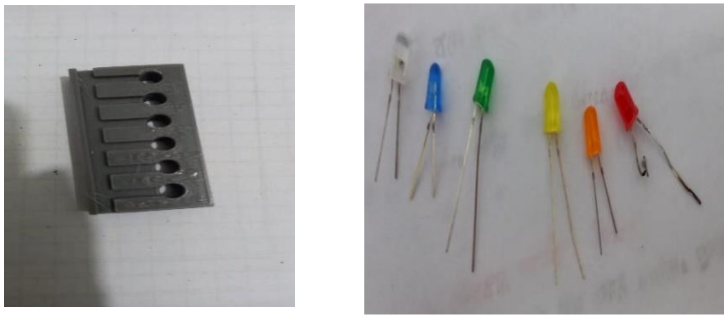


Figure 4.2. Representation of a LED holder and the LEDs.

In the case of phototransistor holder, the dimensions are almost similar except the change in diameter and thickness and it is shown in figure 4.3. The diameter will be lower and have a value of 5.1mm and the thickness will also be like around 6.5mm.

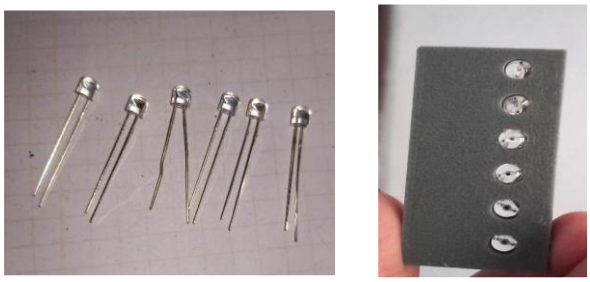


Figure 4.3. Representation of six phototransistors and their array

- **Light Emitting Diode (LED)**

  The LEDs used in this project (violet, blue, green, yellow, orange and red) are key components for producing light at different wavelengths within visible spectrum. Each LED emits light at a specific wavelength, allowing the spectrophotometer to analyze a sample absorption at that wavelength. Different colors which are displayed, enables the device to perform basic spectral analysis, which is essential for identifying and characterizing different chemical solutions. Wavelengths of the LEDs are

  - Violet LED: 390–420 nm
  - Blue LED: 450–490 nm
  - Green LED: 520–560 nm

- Yellow LED: 570–590 nm
- Orange LED: 590–620 nm
- Red LED: 620–750 nm

- **LS-06 light sensors/ phototransistors**

The LS-06 phototransistor is a light sensitive electronic component that detects and converts light into an electrical signal. It functions similarly to a photodiode but offers higher sensitivity due to its amplification capability. They are capable of working in 3-15V DC and they should need to be associated with the 10K ohm resistors. In this project, the LS-06 phototransistor is used to measure the intensity of light transmitted through a sample. The phototransistor's analog signal output is fed into the Arduino's analog input pins, allowing it to capture variations in light intensity corresponding to the absorption characteristics of the sample. Its compact size and cost-effectiveness make it suitable for low cost spectrophotometry applications.

- **Arduino UNO**

The Arduino Uno is a versatile microcontroller board based on the ATmega328P chip. It features 14 digital input/output pins, 6 analog input pins, a 16 MHz quartz crystal and a USB connection for programming and power. The Arduino Uno is widely used in prototyping and DIY electronics projects because of its simplicity and reliability. In this project it has the most significant impact for controlling the LEDs and then take the values of the transistors to gain the light intensity value.

- **Bluetooth module**

The HC-05 Bluetooth module is a popular and low-cost wireless communication device used for establishing serial communication between microcontrollers and Bluetooth-enabled devices like smartphones or computers. It supports both master and slave modes, meaning it can either send or receive data. The module operates on the 2.4 GHz frequency and is often used in projects involving remote control, wireless data transfer or IoT application

- **Resistors**

For minimizing the current flow through LED resistors are being used. Basically, the positive portion of the LED is connected with the digital pins. Now most of the LEDs can't have the capability to be connected with 5V directly. Due to this specific reason six resistors of 100 ohm

are being used. And on the other hand, in the case of phototransistors, 10K ohm resistors are being incorporated for limiting the current that is going to be flown towards the analog pins of the Arduino. As, phototransistors are capable of generating current which is proportional to intensity so in that case if higher current is being generated in that case, before the current will travel towards the Arduino's analog pins a resistance has been introduced so that it doesn't harm the Arduino.

- **Generation of Solution**

In this project we prepared glucose solutions across a wide range of concentration. This concentrations represented different physiological conditions, like hypoglycemia, normal, pre diabetic and diabetic conditions. The concentration range was selected from 50 mg/dL to 250 mg/dL, that re50 mg/dL to 250 mg/dL represents low to high levels of glucose. For example, if we made a solution of 90 mg/dL, it means that we mixed 90 mg of glucose with 100 mL of water to made the solution. Then we take 5mL GOD–POD (Glucose Oxidase–Peroxidase) enzyme in a cuvette and later on add 50 microlitre of prepared glucose solution in the cuvette. Here the ratio is 1:100, which matches the preferred ratio. To maintain consistency a relevant interval was taken for data collection. This allowed for better absorbance change detection and allowed accurate characterization of glucose concentration.



a. b. c.

Figure 4.4. (a)Reagent bottle (b)Glucose bottles (c)Mixed Solution

- **Casing**

The whole process should need to be completed within the darkness. Due to this specific reason the casing has been there in this project which is being made using cardboard. We gained that specific dark area where the whole process can take place. The representation of the casing has been done in figure 4.5.

First of all, we designed the base and the back portions wall of the box. The dimension of the base is like 250mm length, 210mm width and a thickness of 7mm. Leaving a distance of about 2mm from the left and right portion of the base we generated a cut so that right and left walls can be attached in a proper manner. The cut is about 5.5mm and it's about 4mm. A small cut is also being generated of 3mm width and 5mm depth at the front edge so that a sliding front wall can be incorporated.

The backward wall will have a dimension of 250mm length and 190mm height. Apart from that we incorporated a thickness of 7mm. In total it will have three cuts maintaining 2mm from both of the sides and 90mm from the ground edge. The transverse cut is of 5.5mm and its width is around 8mm.

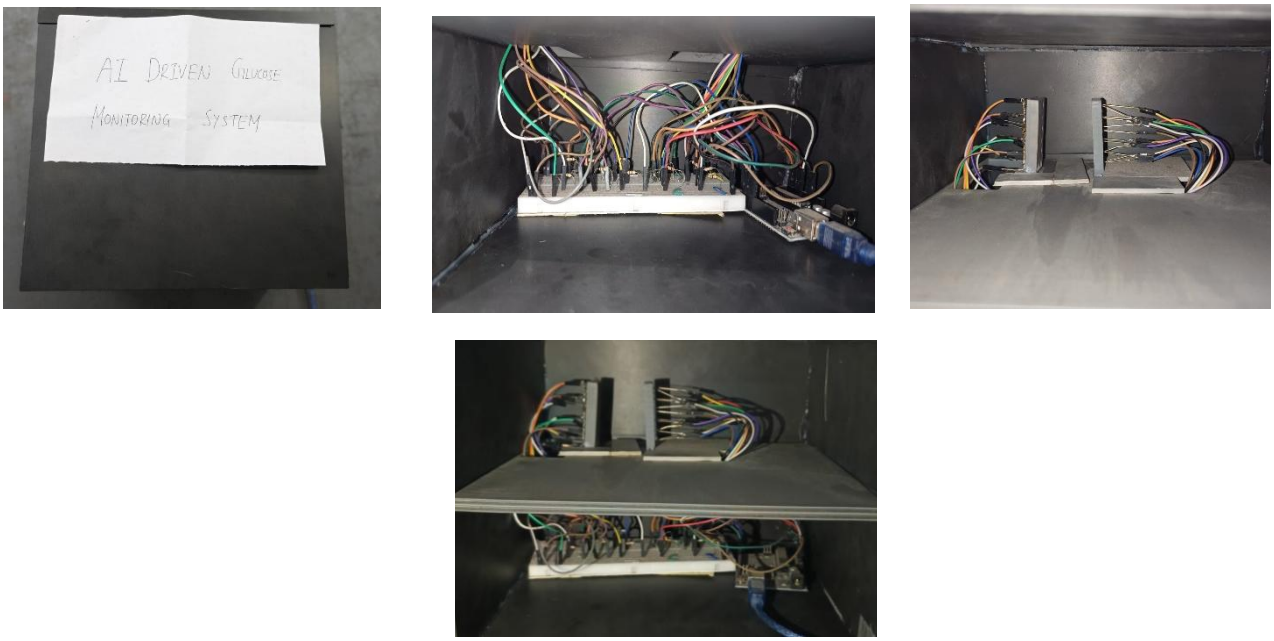


Figure 4.5. Representation of the casing

Considering the left side wall, there will be a length of 203mm and height of 176mm. Extrusion of 5mm is being generated at the ground and the sidewise edge in order to make the proper assemblies. A cut is being generated of 5.5mm depth and a width of 8mm maintaining a distance of 83mm from the edge that is being present in the ground. Similar dimensions are also being incorporated for the right sided wall. Just an extra cut is being there for the Arduino port to be come out. The dimension of that cut is 20 by 15mm.

- **Circuit Diagram**

The circuit diagram, including all components, is developed as shown in figure 4.6

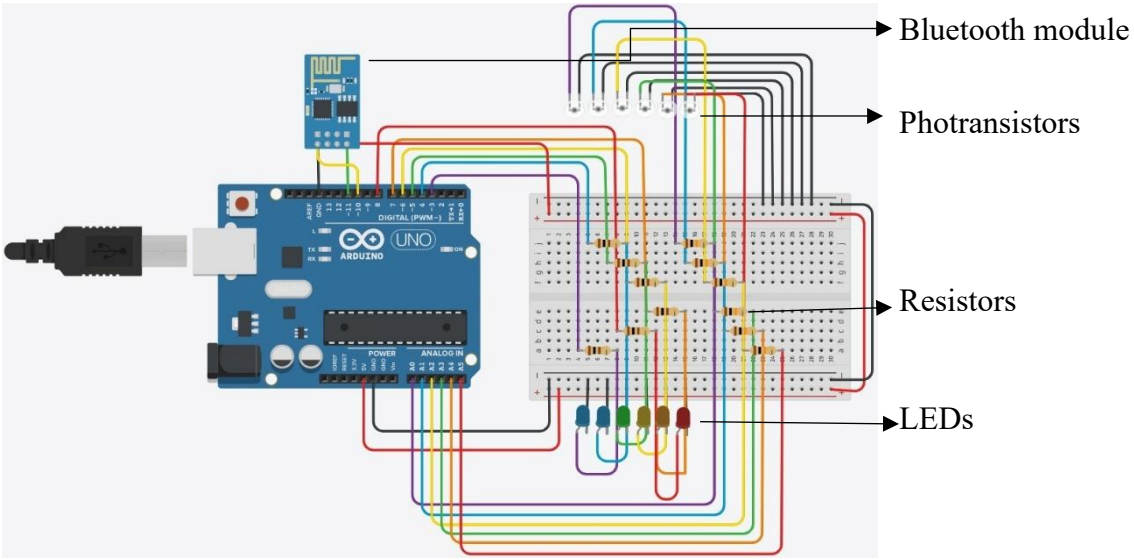


Figure 4.6. Circuit diagram

4.4 Software Setup:

- **Random Forest Regression Model:**

Random forest is a machine learning model that integrates numerous decision trees to reduce the correlation among data. Random Forest reduces the correlation between decision trees by employing a random selection of samples and features. Initially, an equal amount of data is randomly chosen from the training sample data. Furthermore, a subset of the features is chosen at random to construct the decision tree. The utilization of these two forms of randomization results in a decrease in the correlation between each decision tree, hence mitigating the potential mistake caused by overfitting and enhancing the model's accuracy [12]. A schematic of the model is given in figure 4.7.

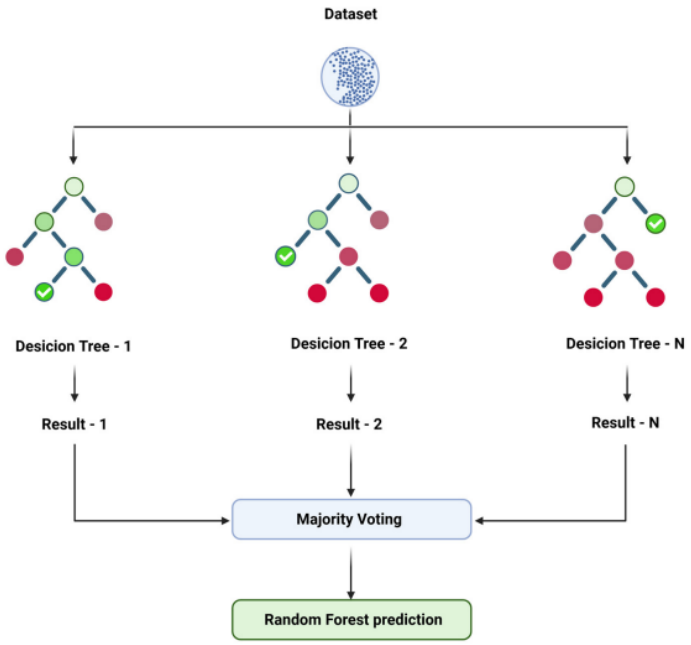


Figure 4.7. Schematic of the Random Forest algorithm [13].

Figure 4.8 illustrates the workflow of the proposed Random Forest-based glucose monitoring system. There are several steps, starting from dataset generation and ending with model exlainability analysis. Each steps contributes to improving the accuracy and reliability of the Random Forest Regression model. The detailed description of each step is represented in next sections.

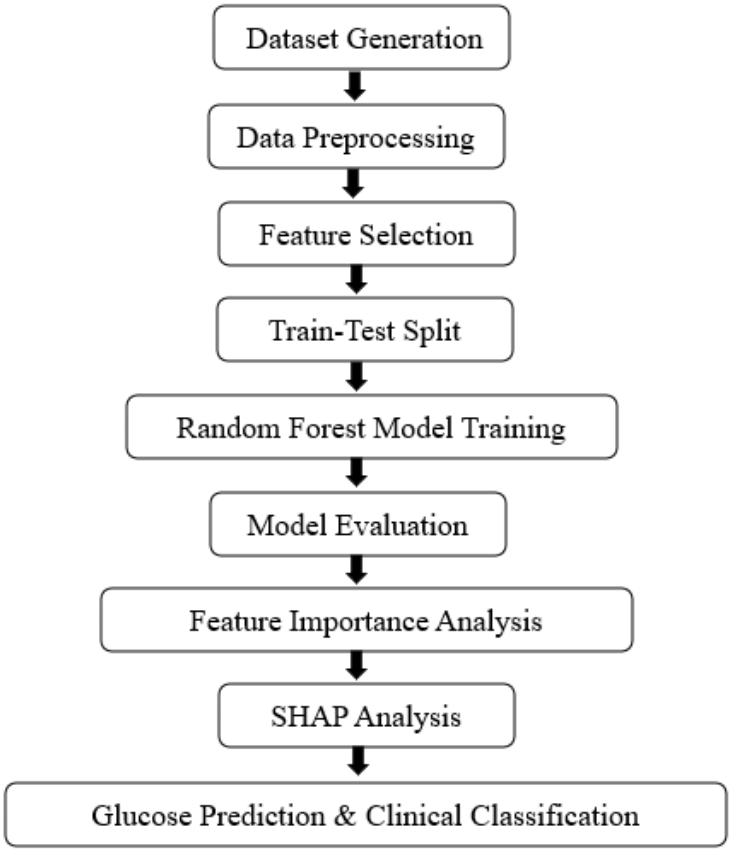


Figure 4.8. Flowchart of the Random Forest-Based Glucose Prediction Model

Step 1: Dataset Generation

The glucose absorbance measurement dataset consists of absorbance responses collected from glucose solutions of different concentrations using multiple LEDs operating at different wavelengths. The dataset was developed using the proposed spectrophotometric system to analyze the relationship between wavelength absorbance and glucose concentration. This dataset is useful for developing machine learning-based predictive models for glucose monitoring and telemedicine applications. Figure 4.9 shows a portion of the generated absorbance measurement dataset.

A	B	C	D	E	F	G
Concentration(per 100 ml)	420	460	520	570	610	630
50	1.12	3.7	12.78	0.51	1.19	1.24
50	3.82	5.67	12.78	0	1.21	1.27
50	1.87	4.67	12.9	1.14	1.28	1.16
50	0	0.9	12.67	1.18	1.25	1.11

Figure 4.9. Representation of the generated absorbance measurement data

Step 2: Data Preprocessing

After dataset generation, preprocessing was performed to improve the quality, consistency and reliability of the data. The dataset was checked for missing, duplicate or unnecessary values, which were removed if present. The data was then organized into a structured format suitable for machine learning analysis. To improve dataset diversity and reduce overfitting, MixUp-based data augmentation was applied. In this process, two random samples were combined using weighted interpolation to generate synthetic samples while preserving the relationship between wavelength absorbance values and glucose concentration.

$$\textbf{MixUp Formula : } x_{new} = \lambda . x_i + (1-\lambda)\, x_j \qquad (4)$$

where, x_i and x_j represent two randomly selected samples from the dataset, λ represents the mixing coefficient and x_{new} represents the generated synthetic sample. Additionally, Gaussian noise was added to the augmented dataset to simulate real-world sensor variability and improve model robustness. The Gaussian noise addition process is represented by the following equation:

$$\textbf{Gaussian Noise Distribution: } X_{noisy} = X + N(0, \sigma^2) \qquad (5)$$

X represents the original dataset, $N(0,\sigma^2)$ represents the Gaussian noise distribution with zero mean and variance σ^2 and X_{noisy} represents the noisebadded dataset.

Step 3: Feature Selection

The absorbance responses collected from six different wavelengths (420 nm, 460 nm, 520 nm, 570 nm, 610 nm, and 630 nm) were organized as input features for machine learning analysis. These wavelength-dependent absorbance values were used to establish the nonlinear relationship between optical response and glucose concentration. Proper feature preparation improved dataset organization, reduced computational complexity, and enabled efficient feature learning by the machine learning model.

Step 4: Train-Test Split

The preprocessed dataset was divided into training and testing subsets to develop and evaluate the machine learning model. Approximately 80% of the dataset was used for model training, while the remaining 20% was reserved for testing and performance evaluation on unseen data.

This strategy ensured unbiased model assessment and improved the generalization capability of the predictive system.

Step 5: Random Forest Model Training

We used a Random Forest Regression model to predict the concentration levels. Random Forest is an ensemble-based supervised machine learning algorithm that integrates multiple decision trees to improve prediction accuracy, minimize variance and reduce overfitting.

During model training, the algorithm learned the complex nonlinear relationship between wavelength absorbance features and glucose concentration values. Hyperparameter optimization was performed manually to improve model performance and stability. The number of estimators (n_estimators) was set to 150 to achieve stable ensemble prediction performance, while the maximum tree depth (max_depth) was limited to 8 to control model complexity and reduce overfitting.

The final glucose prediction generated by the Random Forest model is represented as:

$$\textbf{Prediction Formula :}\ y = \frac{1}{N} \sum_{i=1}^{N} T_i\ (x) \qquad (6)$$

where, $T_i(x)$ represents the prediction obtained from the ith decision tree, N represents the total number of decision trees, and y represents the final predicted glucose concentration.

Step 6: Model Evaluation

Several statistical performance metrices were used to determine the performance of the trained model. The coefficient of determination (R^2 score) was used to measure the correlation between predicted and actual glucose values, while Root Mean Square Error (RMSE) was used to determine the average prediction error. The trained model achieved an R^2 score of **0.960** and an RMSE value of **12.511**, indicating strong predictive capability and low prediction error.

Additionally, 5-fold cross-validation was applied to verify the reliability, consistency, and stability of the model across different subsets of the dataset. The average cross-validation R^2 score obtained was **0.928**, demonstrating satisfactory generalization performance and robustness of the proposed predictive framework.

Step 7: Feature Importance Analysis

Feature importance analysis was performed to quantify the contribution of individual wavelength features to the Random Forest Regression model. The Random Forest algorithm generated feature importance scores based on the contribution of each wavelength feature toward reducing prediction error during decision tree construction.

The analysis demonstrated that the 520 nm wavelength feature exhibited the highest contribution to glucose prediction, indicating strong sensitivity of this wavelength toward glucose concentration variation. This analysis improved the interpretability of the model and identified the most influential wavelength features responsible for prediction accuracy.

Step 8: Glucose Prediction and Clinical Classification

The trained Random Forest Regression model was used to predict glucose concentration values from wavelength absorbance responses collected by the proposed spectrophotometric system. The predicted glucose concentrations were subsequently classified into clinically relevant categories including hypoglycemic, normal, pre-diabetic, and diabetic conditions according to standard glucose concentration thresholds.

This classification stage improved the clinical applicability of the proposed system by enabling automated glucose assessment for healthcare monitoring and telemedicine-based decision support applications.

Step 9: SHAP Analysis

SHAP (SHapley Additive exPlanations) analysis was performed to interpret the contribution of individual wavelength features toward model prediction outcomes. Unlike conventional feature importance analysis, SHAP analysis provides sample-wise interpretability by explaining how each feature positively or negatively influences individual glucose predictions.

The SHAP summary analysis revealed that the 520 nm wavelength feature had the highest impact on model prediction performance, while other wavelength features demonstrated comparatively lower contributions. Features exhibiting larger SHAP value distributions

contributed more significantly to glucose prediction outcomes. This analysis improved the transparency, interpretability, and reliability of the proposed AI-driven glucose monitoring framework and validated the effectiveness of the selected wavelength features for glucose concentration estimation.

- **Mobile Application Development**

A mobile application named **OpTose** was developed for the integration of the AI driven glucose monitoring system with telemedicine and remote healthcare. The icon of the application is given in figure 4.10.



Figure 4.10. Application icon of the developed OpTose mobile application

Kotlin was primarily used for the app logic (actions of the pages) and XML (Extensible Markup Language) was utilized for UI designs (How the app looks). It has total five pages. XML layouts and material design componenets were used to create a user friendly interface. Random Forest model was used for glucose rediction. Each page has their own LinerGradident background and round shaped buttons making sounds while touching utilizing ToneGenerator. An intent based navigation pattern has been followed for connecting the home activity, input activity and result activity. The UI/UX design was developed using the **"Visual Data Insight"** concept to present glucose prediction results in a easy and modern manner. The a has gradient background, custom rogress bar and animated UI components that will imrove user experience. In addition, a Exlainable AI graph was developed using Canvas visualization to display the feature importance. The absorbance values were used as inut to the Random Forest Model for glucose prediction and then those values were classified into hypoglycemic, normal, pre-diabetic and diabetic categories.

Chapter 5

Result & Discussion

5.1 Result & Discussion

Figure 5.1, 5.2 & 5.3 represent the absorbance graph for 130 mg,150 mg and 250 mg of glucose concentration.

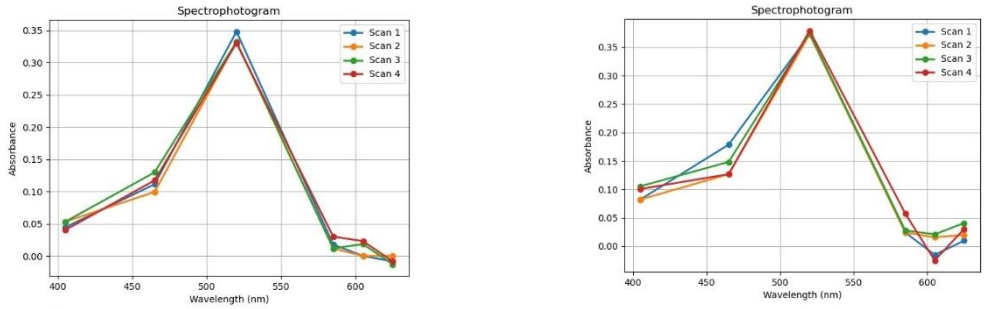


Figure 5.1. Representation of graph for 130mg of glucose concentration

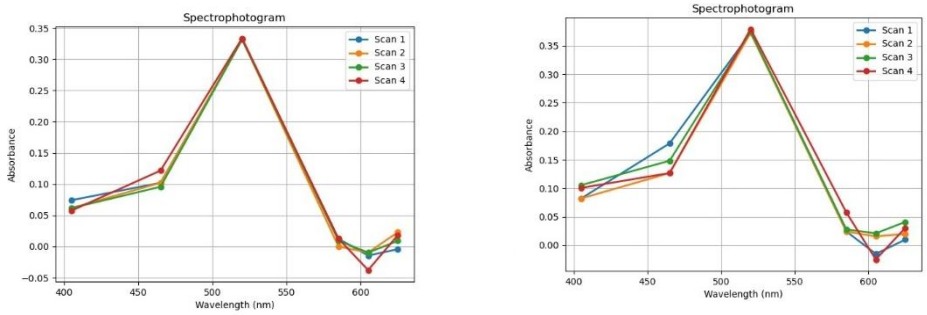


Figure 5.2. Representation of graph for 150mg of glucose concentration

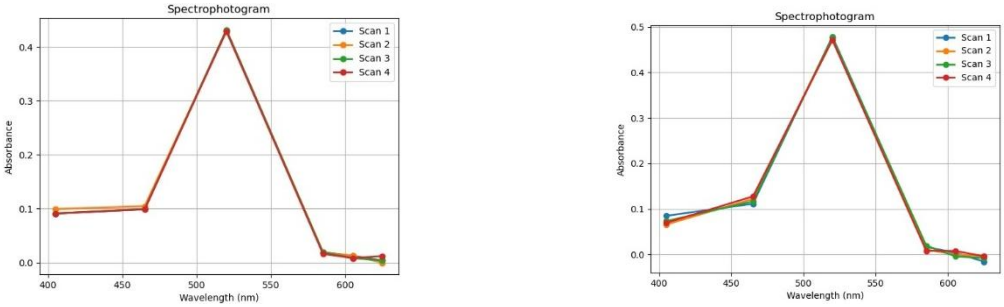


Figure 5.3. Representation of graph for 220mg of glucose concentration

Taken together, figures 5.1–5.3 represent the absorbance responses obtained from the developed spectrophotometric system for glucose concentrations of 130 mg/dL, 150 mg/dL and 220 mg/dL respectively. For each concentration, multiple scans were done to ensure repeatibility stability of the measurements. Figure 5.1 shows the spectrophotometric

absorbance responses for the 130 mg/dL glucose concentration obtained from 2 scans. Bothe the graphs shows similar absorbance pattern. Significance absorbance peak can be observed at near 520nm in both the graphs and the absorbance remain around 0.35 in both cases. Similarly, the lower absorbance peaks are found at higher wavelengths like 610 and 630nm. For 150 mg/dL and 220 mg/dL also, similar repetitive absorbance patterns can also be observed across multiple scans. As we are taking mulitple readings for each concentration, this reduces the chances for any kind of error values. All the measurements were taken in a dark environment, which is critical for minimizing the interference that may happen from outside lights. Overall, the experiment result shows good consistency in the values and the use of repeated measurement, controlled timing, enzymatic specificity and dark environment strengthen the quality and reliability of the dataset. But still some errors may still exist due to the hardware limitations, as it is not a commercial setup.

Figure 5.4 demonstrate the SHAP summary plot, which explains the contribution of different wavelength features to the glucose prediction performance of the Random Forest Regression model.

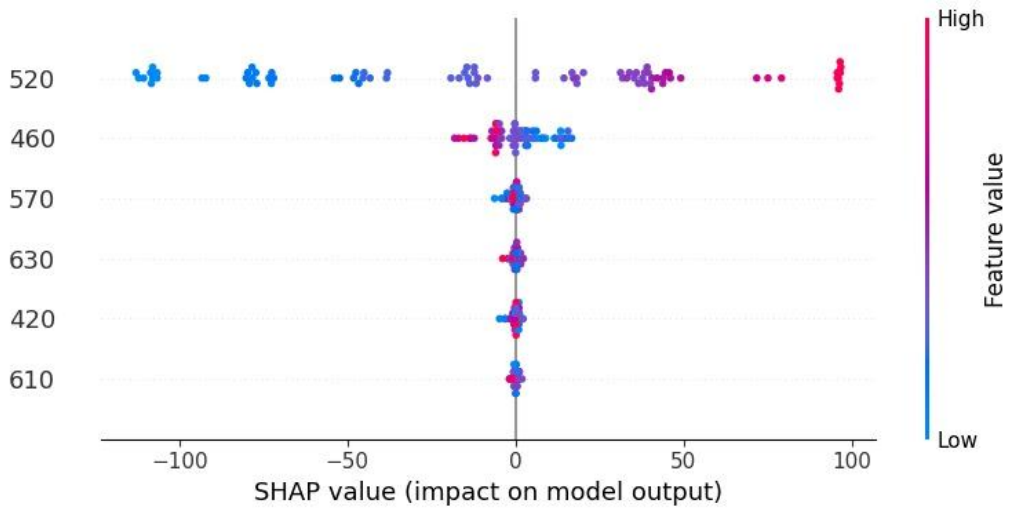


Figure 5.4. SHAP summary plot

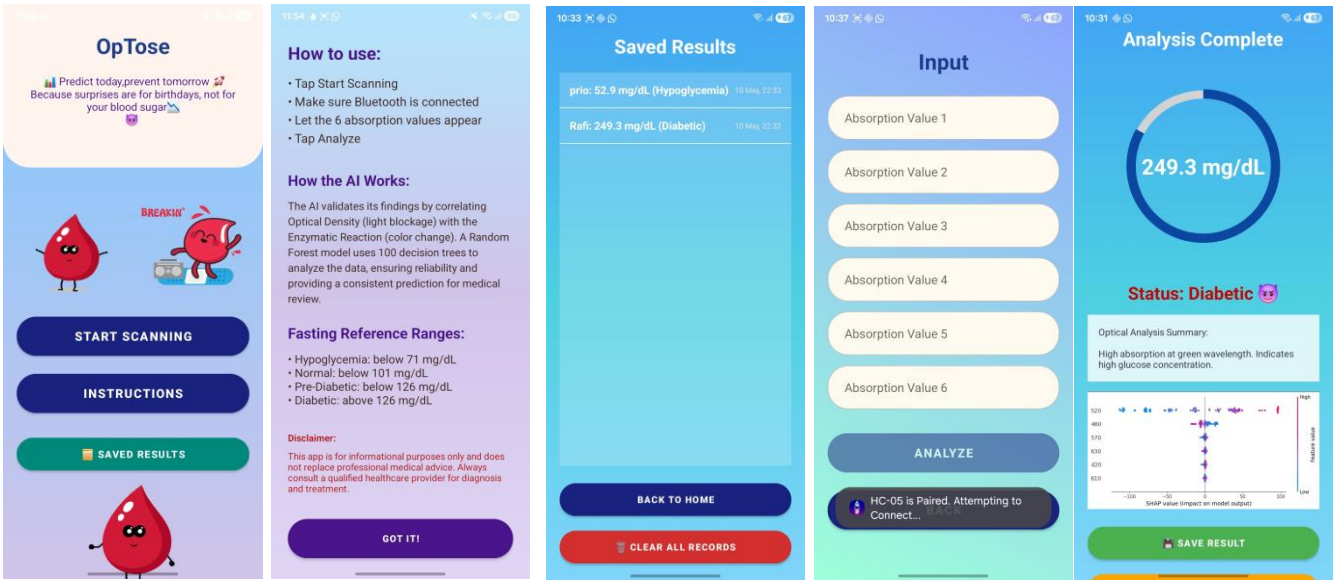
In the figure, the x axis represents the SHAP value, that is the impact of each feature on the model output. Positive SHAP values increase the predicted glucose concentration, while negative SHAP values decrease the prediction result. The y-axis represents the wavelength features used in the model, that is, 520 nm, 460 nm, 570 nm, 630 nm, 420 nm and 610 nm. The

color of the points represents the feature value, here blue indicates lower feature values and ink represents the opposite one. Among all the wavelengths, 520nm shows the highest SHAP value distribution, which indicated that it has the highest influence on glucose concentration prediction. In contrast, other wavelengths have SHAP values clustered near zero, which indicates lower contribution to the model prediction. Positive SHAP values increase the predicted glucose concentration, while negative SHAP values decrease the prediction result. Higher absorbance values at 520 nm generally contribute positively to the predicted glucose concentration, on the other hand, lower values reduce the prediction output.

Figure 5.5 illustrates the user interface of the develoed mobile application. All the interfaces of the application is explained below.

1. Home Interface:

The homepage is the first page showed up when an user opens the application. It has a soft linear gradient background from soft pink to soft blue for better readability, having some animations (moving pictures) for funky visualization. It also has a text box representing basic introduction of our application with a uo-to-down transition via TranslateAnimation class.



(a) (b) (c) (d) (e)

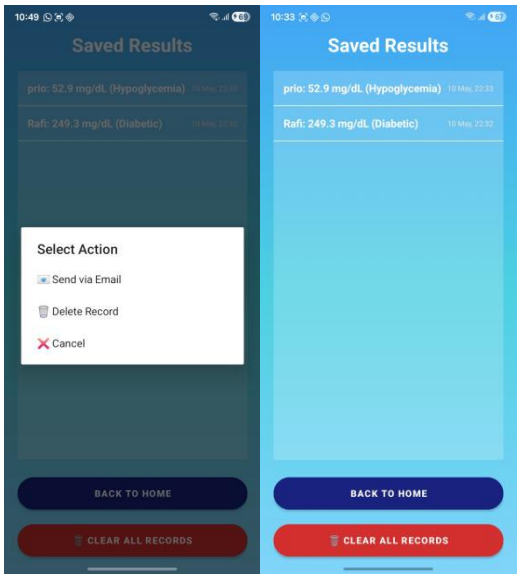
Figure 5.5. The Application pages (a) Home page, (b) Instruction page, (c) History page, (d) Input page and (e) Result page

2. Instruction Interface:

When the instruction button is clicked in the home page, it will take the user to this page. This page includes all the basic instructions on how to use the app and AI interpretations. Navigation is managed using an Intent system, allowing seamless bi-directional transit between the instruction set and the home dashboard.

3. History Interface:

This page stores the result data and patient information (name). It allows for both the manual deletion and all deletion at once and it also allows for sending the information to the patennt through email as shown in figure 5.6. SharePreferance API was used for light-weight and resistant storage of the patient data. The features are done using a SharedPreferences storage system and a ListView with an ArrayAdapter to manage the dynamic data.



(a) (b)

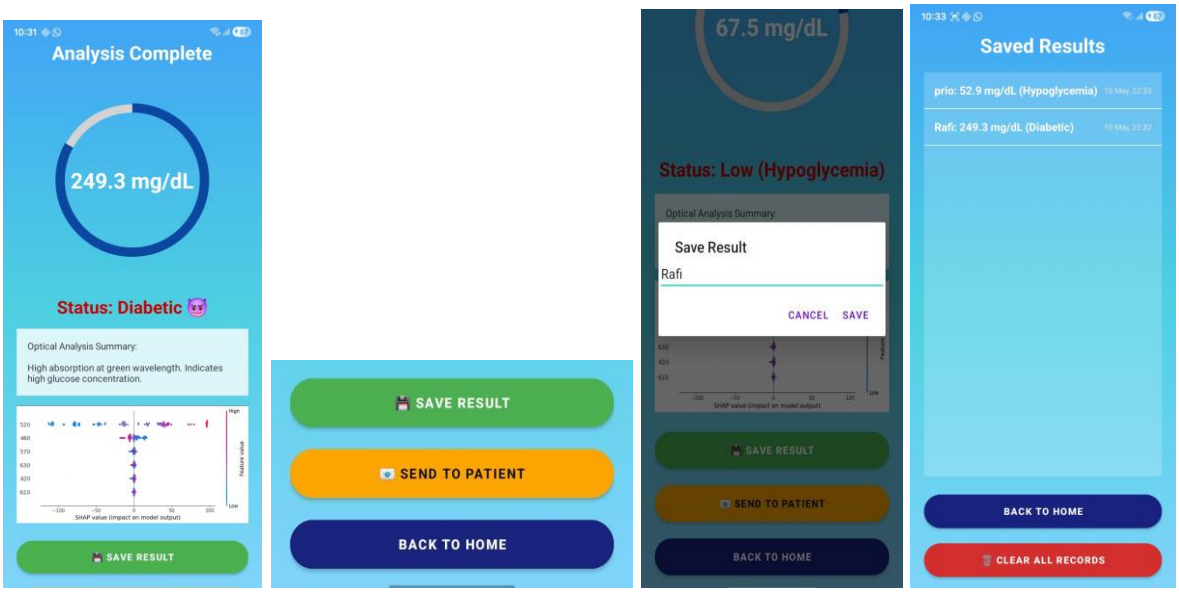
Figure 5.6. (a) Manual Deletion and (b) All deletion at once button

4. Input Interface:

This page creates the bridge between the software and the hardware. Transition was used for input boxes using staggered postDelayed animations and there is a Bluetooth services that allows the six absorption values to come up one by one by itself, when all values are here the analyze button pops up. On triggering the analysis, raw inputs are encapsulated into a DoubleArray and transmitted to the GlucosePredictor engine via an explicit Intent.

5. Result Interface:

It shows the result including glucose concentration, status, optical analysis and SHAP plots for clinical transparency. An animated Gauge Visualization is used to show the level of glucose.All these interfaces are shown in figure 5.7. A save button is also integrated for saving patient information and patient data. Last but not the least the result can be sent to the patent through



(a) (b) (c) (d)

Figure 5.7. (a) Animated gauge visualization, status bar, optical analysis and SHAP plots, (b) Result page buttons, (c) Saving patient information and (d) Saved results

Chapter 6

Conclusion & Future Work

6.1 Conclusion

In this project, we aimed to develop a low-cost, portable and AI- accumulated glucose prediction system by integrating spectrophotometric techniques. We also used enzymatic analysis and telemedicine concepts. A spectrophotometer-based setup has been successfully implemented to measure glucose concentration using the GOD–POD method, based on light absorption principles. Absorbance measurement s were collected from multiple LED wavelengths, including 420 nm, 460 nm, 520 nm, 570 nm, 610 nm, and 630 nm, to analyze the relationship between wavelength response and glucose concentration. The experimental results have been used to create a structured dataset, where multiple values had been added for one concentration level to find the accuracy and stability of the results. The dataset preparation phase plays an important role in ensuring the accuracy and performance of AI models. The collected dataset was then used to train a Random Forest Regression model for glucose concentration prediction. The model demonstrated effective prediction capability, while feature importance and SHAP analyses improved the interpretability of the system by identifying the contribution of different wavelength features. The result shows that 520 nm plays the most significant role in glucose prediction. The project demonstrates strong potential for combining spectrophotometry with AI to develop a cost-effective and practical solution for glucose prediction. Despite its effectiveness, the system may have some limitation in term of spectral resolution and accuracy compared to commercial spectrophotometers due to the use of low cost hardware components, limited dataset size and possible environmental variations during measurement.

6.2 Future Work

In future, the developed system can be improved by collecting a larger dataset to increase the accuracy and reliability of the model. More advance machine learning or deep learning models can be used. At present the hardware setup is not up to the mark. The hardware setup can be further optimized using higher precision components to reduce any kind of noise. Also, the current bluetooth based communication system can be replaced with a wifi module to support

faster data transfer and remote connectivity. In addition, a dedicated website can be developed for cloud data storage and advance telemedicine support.

6.3 Gantt chart

The Gantt chart below outlines a 13 week project timeline for developing the system. It starts with project planning and ends with the complete output. Early weeks focus on core hardware development and later one focuses on software related works like model and app development.

. Table 6.1. Gantt chart of the work

Task \ Week	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10	Week 11	Week 12	Week 13
Project research and topic selection													
Proposal submission													
Proposal acceptance													
Circuit setup and data collection													
Progress report & presentation													
Dataset generation and model training													
Result acquisition and app development													
Final presentation													

Chapter 7

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